TRIVIAL COVER

How to use this handout

Read me first, Detective.

Welcome, Detective! This book is your **case file** for the biggest mystery of our time: machines that seem smart. You will not need a computer, a phone or an app — your tools are paper, pencils, cards and your own sharp eyes. Every chapter hands you missions (we call them **tasks**), strange clues to examine, and empty spaces to fill with what you discover. Real detectives write everything down, so this book belongs to you: write, draw, tick, cross out and try again.

1 READ

Read the mission twice. Underline anything that puzzles you — puzzles are clues.

2 DO

Do the task with your partner, group, class or family, exactly as the chips say.

3 WRITE

Write what you saw in the spaces, then shade your confidence circles honestly.

The work-mode chips

ALONE	your own case — you investigate by yourself
PAIR	two detectives — share one book, take turns reading codes aloud
GROUP	three to four detectives — decide roles before you start
CLASS	the whole class plays together, led by your teacher
HOME	a mission for home — show a grown-up what you discovered

HOW TO READ A MISSION

T6-01 PAIR 10 MIN QUICK WIN

Every mission has a code like **T6-01** — **T6** means this Class 6 handout, and **01** is the mission number, so your teacher can say "open Task T6-08". The **PAIR** chip tells you who to work with, the **10 MIN** chip is your time limit, and **QUICK WIN** marks an easy first success. Chapters 1 and 2 both start with a Quick Win to warm up your detective muscles — nobody fails in this book, because every attempt teaches a clue.

THE QUESTION HABIT — YOUR SUPERPOWER

Three questions make you a sharper detective than any machine: **"How do I know?"** · **"What is missing?"** · **"Who made this?"** Use them in this book, on the internet, in the news, and when a machine gives you an answer that looks too perfect. Detectives are not people who know everything — they are people who **keep asking**.

A note for grown-ups

This handout builds **AI understanding** — how machines learn from data and how humans should judge them — through 20 paper-and-pencil missions. It is fully **unplugged**: no task requires a device, an account, a photo, a voice recording or any personal information. The best help you can give is to ask the three Question-Habit questions, praise careful **reasoning** more than quick right answers, and treat the confidence circles as honest thinking, not scores. A separate teacher pack carries answer notes and rubrics; this book deliberately never prints solutions beside a task.

1

Machines That Seem Smart

Guiding question: When is a machine actually learning — and when is it only following steps?

I WILL LEARN TO · 6.U1 6.U2 6.L2 6.W1 6.T1 STRANDS · U W

Links: Maths — patterns · English — exact instructions

1 Smart machines are everywhere

Every day, machines seem to do clever things. A video app picks the next clip you actually want to watch. A voice assistant answers when you ask about the weather. A map app finds a faster road when traffic jams up. It is easy to think: **the machine is smart, like a person.** But a good detective looks closer before deciding. Under every clever trick there are only two possible explanations: either somebody wrote **exact steps** for the machine to follow, or the machine **learned a pattern from examples**. On this case, you will learn to tell the two apart — and that one skill will change how you see every screen, speaker and gadget around you.

T6-01 Human Robot

QUICK WIN · PAIR · 10 MIN

One of you is the **Robot**, the other is the **Engineer**. The Engineer writes exact steps for the Robot to make a paper plane, or to walk from the door to the chalkboard. Then swap. The Robot must follow the steps **literally** — if the step says "fold the paper", fold it any way you like! When both of you have played both roles, write here:

One step my Robot followed too literally:

What this taught me about giving instructions to machines:

WORD POWER · 1

algorithm

An exact list of steps that tells a machine how to do a task, one step at a time.

Say it three times today — it sticks!

DETECTIVE'S NOTE

The Human Robot game is why programmers and AI designers must be **exact**. A machine never guesses what you **meant** — it only knows what you **said**. When a machine does something surprising, ask: was the algorithm wrong, or was it the data?

2 Rule-followers: same steps, every time

A washing machine washes exactly the same way today as it did last month. A lift goes up and down by fixed rules. A calculator will never get better at adding — it already follows its algorithm perfectly. Machines like these do **automation**: work done by fixed steps, repeated the same way every time. They are useful and fast, but notice something important: they never **improve**. Nobody has to teach them; nobody can. They are rule-followers, and rule-followers are not the same as learners.

T6-02 Rule or Learner?

PAIR · 10 MIN

Sort these eight machines. For each one, write **R** if you think it mostly follows fixed rules, **L** if you think it learns from examples, or **?** if you are honestly not sure. Then write one clue that helped you decide.

Machine	R / L / ?	One clue that helped you
"Washing machine"		
"Lift"		
"Video app that suggests clips"		
"Spam filter in email"		
"Calculator"		
"Traffic signal"		
"Voice assistant"		
"Map app that reroutes you"		

The machine I argued about most with my partner:

WORD POWER · 2

automation

Work done by a machine that repeats fixed steps in the same way every time.

Say it three times today — it sticks!

MYTH BUSTER

MYTH *"AI is a robot."*

TRUTH AI is a **program** — a learner made of data and patterns. It can live inside a phone, a website, a car or a power plant, and it has no body at all. A robot is one place a program can be kept; plenty of robots run with no AI, and plenty of AI runs with no robot.

3 Learners: machines that improve from examples

So how does a video app get good at suggesting clips you like? Nobody wrote a rule for "show Aarav cricket clips". Instead, the app **learned from examples** — millions of them. It noticed which clips people similar to you watched, and it found **patterns** in that. This is the big secret of this whole handout: a learning machine is not given the answer; it is given **examples**, and it works the pattern out by itself.

Here is the strange part: the people who build AI do not write the final rules either. **They choose the examples and the data**, and the machine does the pattern-finding. That is why the examples matter so much — feed a learner one-sided examples, and it learns a one-sided view of the world, exactly like the alien in your next mission.

T6-03 Teach the Alien

GROUP 15 MIN

In your group of four, one person is the **Alien** — smart, but knowing nothing about Earth. The other three must teach the Alien the idea of **mango** using ONLY examples: point at real objects, drawings or the cards your teacher gives you, and say "mango" or "not mango". No describing words allowed — aliens don't speak Human! When the Alien starts guessing correctly, play the twist round: show **only green mangoes**. Then answer:

What wrong idea did our examples put in the Alien's head?

How could we choose better examples next time?

WORD POWER · 3

artificial intelligence (AI)

Technology that lets machines do tasks that seem to need human intelligence.

Say it three times today — it sticks!

WORD POWER · 4

training examples

The examples we show a machine so that it can find patterns by itself.

Say it three times today — it sticks!

DETECTIVE'S NOTE

Machines **predict, match and estimate** — those are the verbs of AI. People **design** the system, **choose the data** and are **responsible** for what it does. Keep those verbs straight and you will never be fooled by a "smart machine" story.

4 Humans vs machines — the fair contest

So is a machine ever really smart? Try a fair contest. Machines are dazzling at some jobs: counting a million numbers without a mistake, noticing a pattern in a million examples, working all night without getting bored. People are dazzling at others: understanding a joke, knowing that your friend is sad from one look, deciding what is **fair**, asking a question nobody has asked before. The contest has no overall winner — and that is exactly why people and machines make a good team, as long as the **people** stay in charge of the judging.

T6-04 Better At / Worse At

ALONE · 10 MIN

Fill the table, then your class votes for the most surprising idea.

Two things PEOPLE do better — and why	Two things MACHINES do better — and why

The class's most surprising idea was...

5 Three ways machines learn — three little stories

Machines learn in more than one way, and each way has a human story you already know. Watch for all three in this book.

STORY 1

Learning with labels

Show it, tell it. Like learning fruit names with flashcards: your teacher shows a card and says "mango". The machine gets examples **with** answers attached — "this photo is a cat", "this one is not". It learns to match new photos to the labels.

STORY 2

Grouping without labels

Sort it your way. Like tidying a mixed toy box with no instructions: you make your own piles — colours here, shapes there. The machine gets examples with **no** answers and must find its own groups, like sorting customers or animals it has never been told about.

STORY 3

Trial and reward

Try it, score it. Like learning to ride a bicycle: wobble, fall, adjust, try again — and a happy feeling keeps you going. The machine tries actions, gets points for good ones, and slowly learns the moves that earn the most reward.

Real AI systems often mix all three. Class 7 will show you what each way is called — for now, the stories are enough.

Self-check

shade one: first circle = not yet · middle = almost · last = got it

- ☐ I can point to a **rule-follower** and a **learner** in my own home, and say which is which ○○○
- ☐ I can explain "learning from examples" in my own words, without the book's help ○○○
- ☐ I can name one thing people do better than machines — and say **why** ○○○

Think & Ink

One machine I use seemed really smart. Now I think it is a … (rule-follower / learner), because …

HOME LINK *take this page home — no screen needed*

T6-05 AI Detective at Home

HOME · 15 MIN

Walk through your **morning routine** like a detective. At each moment below, tick whether a machine with possible AI **may** hide there, and tick whether you think it **learns** from examples or just follows rules. Ask a grown-up if you are stuck — explaining your guess is part of the mission.

Moment in the morning	Possible AI? (✓)	Learns or follows rules?	What data might it use?
"Waking me up (alarm / phone)"			
"Brushing (electric brush / tap)"			
"Breakfast (microwave / kettle)"			
"Checking the day (weather app)"			
"Journey to school (bus / car / map)"			
"Buying something (shop / UPI / ATM)"			

My biggest surprise on this case:

CASE NOTES — WHAT CHANGED IN MY THINKING?

Before this chapter I thought "smart machine" meant one thing. Now I know there are **two kinds**:

MYTH BUSTER

MYTH ***"AI is alive — it has feelings."***

TRUTH A learner finds patterns in numbers; it does not **feel** anything. When a chatbot writes "I am happy to help!", that sentence was predicted from patterns in text people wrote. The **people** who design, choose data and take responsibility have the feelings — and the responsibility.